

2.4 Exact Equations

Given a function $f(x,y)$, its differential is

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy.$$

In this section, we will consider differential equations of the form:

$$M(x,y) dx + N(x,y) dy = 0.$$

The equation is said to be exact if there exists a function $f(x,y)$ such that:

$$M(x,y) = \frac{\partial f}{\partial x} \quad \text{and} \quad N(x,y) = \frac{\partial f}{\partial y}.$$

The DE is then equivalent to $\boxed{df = 0}$.

The general solution is then:

$$\boxed{f(x,y) = C}.$$

ex: ① Consider the DE

$$3x^2 e^{2y} dx + 2x^3 e^{2y} dy = 0.$$

By inspection we can see that the left side is the differential of

$$f(x,y) = x^3 e^{2y} \rightarrow \begin{cases} \frac{\partial f}{\partial x} = 3x^2 e^{2y} \\ \frac{\partial f}{\partial y} = 2x^3 e^{2y} \end{cases}$$

The DE is then :

$$d \left[x^3 e^{2y} \right] = 0$$

$$\Rightarrow \boxed{x^3 e^{2y} = C}$$

We can't expect to solve DEs by inspection.
We will present a systematic way to solve exact equations.

Consider the DE :

$$M(x,y)dx + N(x,y)dy = 0.$$

How can we tell if the equation is exact ?

It is exact if and only if

$$\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}.$$

If it is exact, there is a function $f(x,y)$ such that :

$$\frac{\partial f}{\partial x} = M(x,y) \quad \text{and} \quad \frac{\partial f}{\partial y} = N(x,y).$$

We can find $f(x,y)$ by using partial integration.

Method ①

$$f(x, y) = \underbrace{\int M(x, y) dx}_{y \text{ kept constant}} + \underbrace{g(y)}_{\text{Arbitrary function of } y}$$

$g(y)$ plays the role of the "constant" of integration

Since $\frac{\partial f}{\partial y} = N(x, y)$

then: $\frac{\partial}{\partial y} \int M(x, y) dx + g'(y) = N(x, y)$

$$\Rightarrow g'(y) = N(x, y) - \frac{\partial}{\partial y} \int M(x, y) dx$$

We can integrate to find $g(y)$.

The DE general solution is

$$\boxed{f(x, y) = C}.$$

Method ②

$$f(x, y) = \underbrace{\int M(x, y) dx}_{y \text{ kept constant}} + g(y) \quad (*)$$

$$f(x, y) = \underbrace{\int N(x, y) dy}_{x \text{ kept constant}} + h(x) \quad (**)$$

If $f(x, y)$ is compatible with both
(*) and (**), then:

$$\boxed{f(x, y) = C} \quad \text{is the}$$

general solution of the DE.

You can follow either method ① or ②
to solve an exact DE.

ex: (2) Solve the DE

$$\underbrace{(2xy + 3x^2)}_M dx + \underbrace{(x^2 - 1)}_N dy = 0$$

Let's check if the DE is exact.

$$\frac{\partial M}{\partial y} = 2x$$

> equal, so it is exact.

$$\frac{\partial N}{\partial x} = 2x$$

We will now solve the DE using both methods (1) and (2).

Method ①

$$\begin{aligned} f(x, y) &= \int M \, dx = \int (2xy + 3x^2) \, dx \\ &= x^2 y + x^3 + g(y) \end{aligned}$$

$$\frac{\partial f}{\partial y} = N(x, y)$$

$$\Rightarrow x^2 + 0 + g'(y) = x^2 - 1$$

$$\Rightarrow g'(y) = -1$$

$$\Rightarrow g(y) = \int (-1) \, dy = -y$$

[We don't put a constant in $g(y)$ since there will be one in the final solution.]

General solution : $f(x, y) = C$

$$\Rightarrow \boxed{x^2 y + x^3 - y = C}$$

Method (2)

$$\begin{aligned} f(x, y) &= \int M dx = \int (2xy + 3x^2) dx \\ &= x^2 y + x^3 + g(y) \quad (*) \end{aligned}$$

$$\begin{aligned} f(x, y) &= \int N dy = \int (x^2 - 1) dy \\ &= x^2 y - y + h(x) \quad (**) \end{aligned}$$

We need f to be compatible with both $(*)$ and $(**)$.

$$\Rightarrow f(x, y) = x^2 y + x^3 - y$$

(Just include every term in $(*)$ and $(**)$ only once.)

⚠ Do not write $x^2 y$ twice.

General solution : $x^2 y + x^3 - y = C$

ex: ③ Solve the following DE :

$$\underbrace{(3y \cos(3x) + 2x e^y)}_M dx + \underbrace{(\sin(3x) + x^2 e^y - 4e^{2y})}_N dy = 0.$$

Let's check if the DE is exact.

$$\left. \begin{aligned} \frac{\partial M}{\partial y} &= 3 \cos(3x) + 2x e^y \\ \frac{\partial N}{\partial x} &= 3 \cos(3x) + 2x e^y \end{aligned} \right\} \text{equal, so it is exact.}$$

$$\begin{aligned} f(x, y) &= \int M dx = \int (3y \cos(3x) + 2x e^y) dx \\ &= y \sin(3x) + x^2 e^y + g(y) \end{aligned}$$

$$\frac{\partial f}{\partial y} = N \rightarrow \sin(3x) + x^2 e^y + g'(y)$$
$$\dots = \sin(3x) + x^2 e^y - 4e^{2y}$$

$$\Rightarrow g'(y) = -4e^{2y} \xrightarrow{\text{(integrate)}} g(y) = -2e^{2y}$$

General solution: $y \sin(3x) + x^2 e^y - 2e^{2y} = C$

ex: (4) Solve the following IVP.

$$\begin{cases} \frac{dy}{dx} = \frac{2x - e^y}{xe^y + 2y} \\ y(2) = 0 \end{cases}$$

Let's rewrite the DE in differential form : $(xe^y + 2y) dy = (2x - e^y) dx$

$$\Rightarrow \underbrace{(e^y - 2x)}_M dx + \underbrace{(xe^y + 2y)}_N dy = 0$$

Is it exact ?

$$\frac{\partial M}{\partial y} = e^y$$

$$\frac{\partial N}{\partial x} = e^y$$

) equal, so it is exact.

$$f(x, y) = \int M dx$$

$$= \int (e^y - 2x) dx$$

$$= xe^y - x^2 + g(y)$$

$$\frac{\partial f}{\partial y} = xe^y - 0 + g'(y) = \underbrace{xe^y + 2y}_N$$

$$\Rightarrow g'(y) = 2y$$

$$\Rightarrow g(y) = y^2$$

General solⁿ: $\underbrace{xe^y - x^2 + y^2}_{f(x, y)} = C$

Find C from $y(2) = 0$ $\begin{cases} x=2 \\ y=0 \end{cases}$

$$2e^0 - 2^2 + 0 = c \rightarrow c = -2$$

$$\therefore \boxed{xe^y - x^2 + y^2 = -2}$$

← IVP solution

If a DE is not exact, in some cases we might be able to find an integrating factor (I.F.). Multiplying the DE by the I.F. would turn it into an exact DE.

Consider the DE: $M(x,y) dx + N(x,y) dy = 0$.

Case ① if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the DE is exact

and we can solve it.

Case ② if $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = f(x)$ ← function of x only

then: $e^{\int f(x) dx}$ is an I.F.

Case ③ if $-\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = g(y)$ ← function of y only

then: $e^{\int g(y) dy}$ is an I.F.

Case ④ otherwise, look for another method to solve the DE.

ex: ⑤ Solve : $\underbrace{6xy}_{M} dx + \underbrace{(4y + 9x^2)}_N dy = 0$

$$\frac{\partial M}{\partial y} = 6x$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$\frac{\partial N}{\partial x} = 18x$$

So the DE is not exact.

Can we find an integrating factor ?

Consider : $\frac{1}{N} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{(6x - 18x)}{4y + 9x^2}$

$$= \frac{-12x}{4y + 9x^2}$$

↙
It depends on both x and y .

We can't find an I.F. from this.

Consider :

$$-\frac{1}{M} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = -\frac{(6x - 18x)}{6xy}$$

$$= \frac{12x}{6xy}$$

$$= \frac{2}{y}$$

It depends only on y .

$$\Rightarrow e^{\int \frac{2}{y} dy} = e^{2 \ln y} = e^{\ln y^2} = y^2$$

y^2 is an I.F.

Multiply both sides of the DE to get:

$$\underbrace{6xy^3 dx}_M + \underbrace{(4y^3 + 9x^2y^2) dy}_N = 0$$

Relabel M and N .

The new DE should be exact.

check it.

$$\frac{\partial M}{\partial y} = 18xy^2$$

$$\frac{\partial N}{\partial x} = 18xy^2$$

} equal, so yes.

Solve the exact DE.

$$\begin{aligned} f(x,y) &= \int M dx = \int 6xy^3 dx \\ &= 3x^2y^3 + g(y) \end{aligned}$$

$$\frac{\partial f}{\partial y} = N \rightarrow 9x^2y^2 + g'(y) = 4y^3 + 9x^2y^2$$

$$\rightarrow g'(y) = 4y^3$$

$$\rightarrow g(y) = y^4$$

General solution:

$$\underbrace{3x^2y^3 + y^4}_{f(x,y)} = C$$